Snowflake Project – Data Engineering & SQL Solution

Date	28 October 2026
Team ID	EN25CS3010016
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	https://github.com/Aanshi-00/Global_Olympic_Sports_Performance_Analysis
Data Source(s) Link	Institution-provided Olympic datasets: alltime_country_rankings.csv, athletes.csv, games_summary.csv, medal_results.csv.
Database	Not applicable — CSV files connected directly to Tableau.
Warehouse	Not applicable — no Snowflake warehouse used.
Tables / Views Created	1. `alltime_country_rankings.csv` — country medal rankings. 2. `athletes.csv` — athlete details and participation. 3. `games_summary.csv` — Olympic games information. 4. `medal_results.csv` — medal results and associated details.
Key Transformations / Calculations	1. Configured relationships between the four datasets. 2. Created the `Unique Athletes` calculated field. 3. Configured Year, Season and Sport filters. 4. Applied aggregations and measures for medal comparisons, trends and regional analysis.

KPIs created	1. Total Olympic medals by country. 2. Gold, silver and bronze medal comparison. 3. Medal trends by year. 4. Unique athlete distribution by gender. 5. Athlete participation by sport. 6. Average medal efficiency by region. 7. Average medal count by investment category. 8. Medal performance by host-country category.
Key Insights	1. Country rankings help compare overall Olympic medal performance. 2. Medal-category charts show differences in gold, silver and bronze achievements. 3. Year-wise trends help explore changes in medal performance over time. 4. Athlete and regional charts reveal participation patterns and differences across categories. 5. Investment-category comparisons show associations in the supplied data, not proof of causation.

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	No — Tableau CSV connections used
2	Data is cleaned, transformed, and prepared appropriately	Yes — basic Tableau data preparation performed
3	Required tables, views, and data relationships are correctly implemented	Yes — Tableau relationships configured; no Snowflake tables/views
4	Required SQL logic, calculations, and KPIs are correctly implemented	N/A — Tableau calculations and measures used instead of SQL
5	Data and solution outputs are validated against the project requirements	Yes — dashboard filters and chart behavior tested
6	Solution is properly organized, documented, and functional	In progress — complete after documentation and remaining project work are finalized

Additional Notes / Comments:

The project, Global Olympic Sports Performance Analysis, was developed using Tableau and four institution-provided CSV datasets. The workbook contains eight visualizations, three dashboards and a four-point Tableau Story. Data relationships, interactive filters and the Unique Athletes calculated field support the analysis. Snowflake database objects and SQL scripts are not part of the current implementation. Further work includes finalizing project documentation and the planned Flask web integration.